

Recursive Macroeconomic Theory

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Complete Solutions

Chapter 6 — Search and Unemployment

All twenty-seven end-of-chapter exercises, 6.1–6.27 (pp. 204–222)

Each exercise has a short restatement of the question (page number given; see the book for the full wording), a step-by-step solution with no skipped algebra, and a verification. Numerical answers come from the companion notebook `Resolucao/ljungqvist_sargent/ls_ch06.ipynb`, which has one section per exercise and checks every number by an independent route — the reservation-wage equation against value-function iteration on the whole function, exact evaluation of cutoff rules, exhaustive policy enumeration, Howard iteration, or Monte Carlo — and aborts on any mismatch. The two exercises that ask for Matlab (6.18, 6.25) are done in Python.

Where the book gives no numbers we choose standard ones and state them. Exercises are grouped by **topic**, not by number. Blue boxes are odd-numbered exercises, purple boxes even.

Ljungqvist–Sargent is supplementary reading and lies outside the course syllabus; this file is for depth, not for the exam.

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Chapter 6 Search and Unemployment

McCall's model (section 6.3). An unemployed worker draws one offer $w \sim F$, $F(0) = 0$, $F(B) = 1$, each period; accepting pays w forever, rejecting pays c today and a fresh draw tomorrow:

$$v(w) = \max \left\{ \frac{w}{1-\beta}, c + \beta \int_0^B v(w') dF(w') \right\}. \quad (6.3.1)$$

Accepting is worth more for higher w , rejecting is a constant Q , so the policy is a *reservation wage* $\bar{w} = (1-\beta)Q$, which solves

$$\bar{w} - c = \frac{\beta}{1-\beta} \int_{\bar{w}}^B (w' - \bar{w}) dF(w') \equiv h(\bar{w}) \quad (6.3.3)$$

or, after integrating by parts, $\bar{w} - c = \beta(Ew - c) + \beta \int_0^{\bar{w}} F(w') dw'$ (6.3.5).

The reduction used throughout. If rejecting is worth Q and accepting is worth $\bar{w}$ -increasing $a(w)$ with $a(\bar{w}) = Q$, then $\int \max\{a(w'), Q\} dF = Q + \int_{\bar{w}} [a(w') - a(\bar{w})] dF$. With $a(w) = w/(1-\beta)$ this is $Q + \frac{1}{1-\beta} \int_{\bar{w}}^B (w' - \bar{w}) dF(w')$. We write $G_F(x) \equiv \int_x^B (w' - x) dF(w') = \int_x^B [1 - F(w')] dw'$ for this option-value integral.

Mean-preserving spreads (section 6.2). F_1 is an MPS of F_2 if (i) the means are equal and (iii) $\int_0^y [F_1 - F_2] d\theta \geq 0$ for all y . Because $g(s) = \int_0^s F$ enters (6.3.5), an MPS raises $\bar{w}$ (section 6.3.2): the worker holds an option and only the right tail matters.

Also used: quits and firing (6.3.3, 6.3.5), waiting times $1/(1-F(\bar{w}))$ (6.3.4), Neal's career choice model (section 6.5, Bellman equation (6.5.1), cutoffs $\theta, \bar{\epsilon}$), and Howard's policy improvement (chapter 3).

Variations on McCall's Bellman Equation

Exercise 6.1 Being Unemployed with a Chance of an Offer

Question

(p. 204) Each period, with probability $\phi \in (0,1)$ the unemployed worker gets no offer (an offer of zero forever); with probability $1-\phi$ she draws $w \sim F$. Income $y_t = w$ employed, c unemployed; jobs last forever. Write the Bellman equation for $v(w)$, the value of an unemployed worker with offer w in hand.

Solution:

Bellman equation. Next period's offer is 0 with probability ϕ and a draw from F with

probability $1 - \phi$:

$$v(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \left[\phi v(0) + (1 - \phi) \int_0^B v(w') dF(w') \right] \right\}$$

Reservation wage. The second argument, Q , does not depend on w ; the first rises with w . So $v(w) = \max\{w/(1 - \beta), Q\}$ and $\bar{w} = (1 - \beta)Q$. The zero offer is rejected whenever $Q > 0$ (true for $c > 0$), so $v(0) = Q$. Using the reduction of the overview, $\int v dF = Q + \frac{1}{1 - \beta} G_F(\bar{w})$, hence

$$Q = c + \beta \phi Q + \beta(1 - \phi) \left[Q + \frac{G_F(\bar{w})}{1 - \beta} \right] \implies (1 - \beta)Q = c + \frac{\beta(1 - \phi)}{1 - \beta} G_F(\bar{w}),$$

and with $(1 - \beta)Q = \bar{w}$:

$$\bar{w} - c = \frac{\beta(1 - \phi)}{1 - \beta} \int_{\bar{w}}^B (w' - \bar{w}) dF(w')$$

This is (6.3.3) with the option value scaled by $1 - \phi$. The right side falls with ϕ and the left side rises with $\bar{w}$, so a higher chance of no offer lowers the reservation wage. The exit probability is $(1 - \phi)[1 - F(\bar{w})]$ per period and the mean duration is its inverse (section 6.3.4).

Numerical exercise — see the companion code

F uniform on $[0, 10]$ (200-point grid), $c = 1$, $\beta = .95$:

ϕ	0	.3	.6
$\bar{w}$ (equation = VFI with an explicit zero offer)	7.4037	6.9970	6.2744
mean duration $1/[(1 - \phi)(1 - F(\bar{w}))]$	3.846	4.762	6.667
simulated (20 000 spells)	3.822	4.764	6.725

Exercise 6.2 Two Offers per Period

Question

(pp. 206–207) The unemployed worker draws *two* i.i.d. offers from F each period; $v(w)$ is the value with best offer w in hand. (a) Bellman equation. (b) Prove that the reservation wage is higher than with one offer per period.

Solution:

(a) The best of two independent draws has c.d.f. $\Pr\{\max(w_1, w_2) \leq w\} = F(w)^2$:

$$v(w) = \max \left\{ \frac{w}{1-\beta}, c + \beta \int_0^B v(w') dF^2(w') \right\}$$

As in McCall, $\bar{w}_2 = (1-\beta)Q_2$ solves $\bar{w}_2 - c = h_2(\bar{w}_2)$ with $h_k(x) \equiv \frac{\beta}{1-\beta} \int_x^B (w' - x) dF^k(w')$.

(b) Integrating by parts, $\int_x^B (w' - x) dF^k(w') = \left[(w' - x)(F^k(w') - 1) \right]_x^B + \int_x^B [1 - F^k(w')] dw' = \int_x^B [1 - F^k(w')] dw'$, so

$$h_2(x) - h_1(x) = \frac{\beta}{1-\beta} \int_x^B F(w')[1 - F(w')] dw' \geq 0,$$

strictly for $x < B$ whenever $0 < F < 1$ on a set of positive length in (x, B) (true for any nondegenerate F once x lies below the top of the support). Let $\psi_k(x) = x - c - h_k(x)$. Each ψ_k is strictly increasing ($h'_k = -\frac{\beta}{1-\beta}[1 - F^k] \leq 0$), and $\bar{w}_k$ is its root. With $c < B$ we have $\bar{w}_1 < B$ and

$$\psi_2(\bar{w}_1) = \underbrace{\psi_1(\bar{w}_1)}_{=0} - [h_2(\bar{w}_1) - h_1(\bar{w}_1)] < 0 \implies \boxed{\bar{w}_2 > \bar{w}_1}.$$

A second offer is a first-order-dominating improvement of the offer distribution: every draw is at least as good, so waiting is worth more and the worker becomes choosier. (If $c \geq B$ nobody ever accepts and both "reservation wages" equal c .)

Numerical exercise — see the companion code

F uniform on $[0, 10]$, $\beta = .95$. $\bar{w}_1$ vs. $\bar{w}_2$: $c = 0$: $7.2395 < 7.8866$; $c = 1$: $7.4037 < 8.0119$; $c = 3$: $7.7613 < 8.2824$; $c = 6$: $8.4079 < 8.7674$. $\bar{w}_2$ is reproduced by VFI on the pair of offers (w_1, w_2) , which never uses F^2 , and the best grid cutoff (exact policy evaluation) is the first grid point above $\bar{w}_2$.

Exercise 6.3 A Random Number of Offers per Period

Question

(p. 207) Each period the worker gets n offers with probability π_n , $n = 1, \dots, N$, i.i.d. over time; offers i.i.d. from F . Formulate the Bellman equation for $v(w)$, the value with best offer w in hand.

Solution:

Conditional on n the best offer has c.d.f. F^n ; unconditionally it has $H(w) =$

$\sum_{n=1}^N \pi_n F(w)^n$. Hence

$$v(w) = \max \left\{ \frac{w}{1-\beta}, c + \beta \sum_{n=1}^N \pi_n \int_0^B v(w') dF^n(w') \right\} = \max \left\{ \frac{w}{1-\beta}, c + \beta \int_0^B v dH \right\}$$

It is McCall with offer distribution H , so $\bar{w} - c = \frac{\beta}{1-\beta} \int_{\bar{w}}^B (w' - \bar{w}) dH(w')$. Since $F^N \leq H \leq F$, the argument of 6.2(b) places $\bar{w}$ between the one-offer and the N -offer reservation wages.

Numerical exercise — see the companion code

$\pi = (.2, .5, .3)$ for $n = 1, 2, 3$, F uniform on $[0, 10]$, $c = 1$, $\beta = .95$: $\bar{w} = 8.0368$ (equation, VFI and the best cutoff agree), between 7.4037 ($n \equiv 1$) and 8.3044 ($n \equiv 3$).

Exercise 6.4 Cyclical Fluctuations in the Number of Job Offers

Question

(p. 207) As in 6.3, but n follows a Markov chain, $\Pr\{n_{t+1} = m \mid n_t = n\} = \pi_{mn}$, $\sum_m \pi_{mn} = 1$. (a) Bellman equation for $v(w, n)$, the value with n offers this period, best w . (b) Show that the optimal policy is a reservation wage $\bar{w}(n)$.

Solution:

(a) Today's n matters only as a predictor of tomorrow's number of offers:

$$v(w, n) = \max \left\{ \frac{w}{1-\beta}, c + \beta \sum_{m=1}^N \pi_{mn} \int_0^B v(w', m) dF^m(w') \right\}$$

(Note the book's index order: the first subscript of π_{mn} is *next* period's state, so the columns of $[\pi_{mn}]$ sum to one.)

(b) Call the second argument $Q(n)$. It depends on n but not on w , while $w/(1-\beta)$ is strictly increasing in w . So for each n the worker accepts iff $w \geq \bar{w}(n) \equiv (1-\beta)Q(n)$. The N numbers $Q(n)$ solve a finite system: by the reduction of the overview, $\int v(w', m) dF^m = Q(m) + \frac{1}{1-\beta} \int_{\bar{w}(m)}^B (w' - \bar{w}(m)) dF^m(w')$, so

$$Q(n) = c + \beta \sum_m \pi_{mn} \left[Q(m) + \frac{1}{1-\beta} \int_{(1-\beta)Q(m)}^B (w' - (1-\beta)Q(m)) dF^m(w') \right],$$

a contraction in $(Q(1), \dots, Q(N))$ with modulus β . When the chain is persistent, many offers today forecast many offers tomorrow, raising $Q(n)$ and hence $\bar{w}(n)$ with n : reservation wages are procyclical.

Numerical exercise — see the companion code

$N = 3$, $\Pr(m \mid n)$ with rows $(.8, .15, .05)$, $(.1, .8, .1)$, $(.05, .15, .8)$; F uniform on $[0, 10]$, $c = 1$, $\beta = .95$: $\bar{w}(1), \bar{w}(2), \bar{w}(3) = 7.7076, 7.9783, 8.1637$. VFI on the full (w, n) grid gives the same numbers, and its accept set is an upper interval in w for each n .

Exercise 6.5 Choosing the Number of Offers**Question**

(p. 208) At cost $k(n)$, increasing in n , the worker receives n offers from F this period; n is chosen at the start of the period. Income: w employed and not searching, $w - k(n)$ in the first period of a job, $c - k(n)$ when unemployed. Formulate the Bellman equation for Q , the value before choosing n .

Solution:

Given n , the best offer w has c.d.f. F^n . Accepting yields $w - k(n)$ today and w forever after, $\frac{w}{1-\beta} - k(n)$; rejecting yields $c - k(n) + \beta Q$. The cost is sunk in both branches, so

$$Q = \max_{n \geq 1} \left\{ -k(n) + \int_0^B \max \left\{ \frac{w}{1-\beta}, c + \beta Q \right\} dF^n(w) \right\}$$

Policy. Given Q , the acceptance rule is a reservation wage $\bar{w} = (1-\beta)(c + \beta Q)$ that does not depend on the n chosen; by stationarity the same n^* is chosen every period of the spell. By the reduction of the overview, $\int \max\{\cdot\} dF^n = c + \beta Q + \frac{1}{1-\beta} \int_{\bar{w}}^B [1 - F^n(w)] dw$, so the gain from one more offer is

$$\Delta(n) = \frac{1}{1-\beta} \int_{\bar{w}}^B F(w)^n [1 - F(w)] dw,$$

decreasing in n . The optimal n^* is the largest n with $\Delta(n-1) \geq k(n) - k(n-1)$.

Numerical exercise — see the companion code

$k(n) = 2n$, F uniform on $[0, 10]$, $c = 1$, $\beta = .95$: $n^* = 6$, $Q = 164.8023$, $\bar{w} = 7.8781$. Iterating on Q and evaluating exactly every stationary rule "(solicit n , accept iff $w \geq x$)" for $n \leq 15$ and every grid x give the same optimum.

Exercise 6.7 Variable Labor Supply

Question

(p. 209) Maximise $E \sum \beta^t u(y_t, l_t)$ with $y_t = c$ if unemployed and $y_t = w(1 - l_t)$ if employed at w (leisure $l_t \in [0, 1]$). Analyse the problem; show the reservation-wage property and that hours are the same every period.

Solution:

Assume u increasing in both arguments. An unemployed worker does not work, $l = 1$. Once employed at w the worker faces the same static problem every period (no savings, w fixed, nothing carried over), so

$$u^*(w) \equiv \max_{0 \leq l \leq 1} u(w(1 - l), l), \quad l_t = l^*(w) \text{ for all } t,$$

and hours $1 - l^*(w)$ are constant over the job. The Bellman equation is

$$v(w) = \max \left\{ \frac{u^*(w)}{1 - \beta}, u(c, 1) + \beta \int v(w') dF(w') \right\}$$

Reservation wage. u^* is nondecreasing: for $w' > w$, $u^*(w') \geq u(w'(1 - l^*(w)), l^*(w)) \geq u(w(1 - l^*(w)), l^*(w)) = u^*(w)$. The accept value is nondecreasing in w and the reject value is constant, so accept iff $w \geq \bar{w}$ with

$$u^*(\bar{w}) - u(c, 1) = \frac{\beta}{1 - \beta} \int_{\bar{w}}^B [u^*(w') - u^*(\bar{w})] dF(w'),$$

(6.3.3) with wages replaced by the indirect utilities they buy. Hours can differ *across* jobs (l^* depends on w through income and substitution effects) but not over time within a job.

Numerical exercise — see the companion code

$u(y, l) = 2\sqrt{y} - 1/l$, $c = 1$, F uniform on $[0, 10]$, $\beta = .95$. $l^*(w)$ from the FOC $1/l^2 = \sqrt{w}/\sqrt{1 - l}$ agrees with direct maximisation to 10^{-5} . $\bar{w} = 7.9348$ (equation = VFI cutoff); hours 0.4992 at $\bar{w}$ and 0.5219 at $w = 10$.

Exercise 6.8 Wage Growth Rate and the Reservation Wage

Question

(pp. 209–210) An accepted offer pays $w_t = \phi^t w$ after t periods of tenure, $\phi > 1$, $\phi\beta < 1$; unemployed get c . Write the Bellman equation. Argue that if $\phi_1 > \phi_2$ the economy with ϕ_1 has the smaller reservation wage.

Solution:

Accepting w is worth $\sum_t \beta^t \phi^t w = \frac{w}{1-\beta\phi}$ (finite because $\beta\phi < 1$):

$$v(w) = \max \left\{ \frac{w}{1-\beta\phi}, c + \beta \int v(w') dF(w') \right\}$$

The reservation wage satisfies $\frac{\bar{w}}{1-\beta\phi} = Q = c + \beta \int v dF$, and $\int v dF = Q + \frac{1}{1-\beta\phi} G_F(\bar{w})$. So

$$\frac{\bar{w}}{1-\beta\phi} = c + \frac{\beta\bar{w}}{1-\beta\phi} + \frac{\beta G_F(\bar{w})}{1-\beta\phi} \implies (1-\beta)\bar{w} = (1-\beta\phi)c + \beta \int_{\bar{w}}^B (w' - \bar{w}) dF(w')$$

Let $\Psi(\bar{w}, \phi) = (1-\beta)\bar{w} - (1-\beta\phi)c - \beta G_F(\bar{w})$. Then $\Psi_{\bar{w}} = 1 - \beta + \beta[1 - F(\bar{w})] = 1 - \beta F(\bar{w}) > 0$ and $\Psi_{\phi} = \beta c$, so

$$\frac{d\bar{w}}{d\phi} = -\frac{\beta c}{1 - \beta F(\bar{w})} < 0 \quad \text{for } c > 0.$$

Hence $\phi_1 > \phi_2 \implies \bar{w}(\phi_1) < \bar{w}(\phi_2)$. Faster wage growth multiplies the value of *every* job by the same factor $1/(1-\beta\phi)$ but leaves the flow c that is given up by accepting unchanged, so waiting becomes relatively less attractive.

Caveat. With $c = 0$ the problem is homogeneous: dividing the Bellman equation by $1/(1-\beta\phi)$ shows $\bar{w}$ is independent of ϕ . The claim is strict only for $c > 0$.

Numerical exercise — see the companion code

F uniform on $[0, 10]$, $\beta = .95$. $c = 1$: $\bar{w} = 7.4037, 7.3403, 7.2780$ for $\phi = 1, 1.02, 1.04$. $c = 0$: $\bar{w} = 7.2395$ for all three. Equation and VFI agree to 10^{-8} .

Exercise 6.12 Search and Financial Income

Question

(pp. 211–212) Every agent also gets financial income ϵ_t , i.i.d. with c.d.f. G , independent of offers. Employed at w : $y_t = w + \phi\epsilon_t$, $0 < \phi < 1$; unemployed: $y_t = c + \epsilon_t$; $1 > \Pr\{w \geq c + (1-\phi)\epsilon\} > 0$. Write the Bellman equation and show that the reservation wage increases with the level of financial income.

Solution:

The state is the pair (w, ϵ) observed this period. Let $\mu = E\epsilon$. Accepting pays $w + \phi\epsilon$ today and $w + \phi\epsilon_{t+s}$ afterwards, worth $\frac{w}{1-\beta} + \phi\epsilon + \frac{\beta\phi\mu}{1-\beta}$:

$$v(w, \epsilon) = \max \left\{ \frac{w}{1-\beta} + \phi\epsilon + \frac{\beta\phi\mu}{1-\beta}, c + \epsilon + \beta \iint v(w', \epsilon') dF(w') dG(\epsilon') \right\}$$

Reduction. Every stream contains $\phi\epsilon_t$ whatever the worker does; subtract its value.

Define $\tilde{v}(w, \epsilon) = v(w, \epsilon) - \phi\epsilon - \frac{\beta\phi\mu}{1-\beta}$. Then $Ev = E\tilde{v} + \phi\mu + \frac{\beta\phi\mu}{1-\beta} = E\tilde{v} + \frac{\phi\mu}{1-\beta}$ and the reject branch becomes $c + \epsilon + \beta E\tilde{v} + \frac{\beta\phi\mu}{1-\beta} - \phi\epsilon - \frac{\beta\phi\mu}{1-\beta}$:

$$\tilde{v}(w, \epsilon) = \max \left\{ \frac{w}{1-\beta}, \tilde{c}(\epsilon) + \beta U \right\}, \quad \tilde{c}(\epsilon) \equiv c + (1-\phi)\epsilon, \quad U \equiv E\tilde{v}.$$

This is McCall with an i.i.d. compensation $\tilde{c}(\epsilon)$: an employed worker has less time to collect financial income, so unemployment is subsidised by $(1-\phi)\epsilon$. The assumption $1 > \Pr\{w \geq \tilde{c}(\epsilon)\} > 0$ says that the job's flow beats unemployment's flow sometimes but not always, so the problem is not trivial. The policy is

$$\bar{w}(\epsilon) = (1-\beta) \left[c + (1-\phi)\epsilon + \beta U \right], \quad U = \iint \max \left\{ \frac{w}{1-\beta}, \tilde{c}(\epsilon) + \beta U \right\} dF dG.$$

- (i) *Current financial income*: $\partial \bar{w} / \partial \epsilon = (1-\beta)(1-\phi) > 0$.
- (ii) *Level (mean) of financial income*: shift ϵ by m . Differentiating the equation for U , with ρ the probability of rejecting, $dU = \rho[(1-\phi)dm + \beta dU]$, so $dU/dm = \rho(1-\phi)/(1-\beta\rho) > 0$, and $d\bar{w}(\epsilon)/dm = (1-\beta)(1-\phi)[1 + \beta\rho/(1-\beta\rho)] = (1-\beta)(1-\phi)/(1-\beta\rho) > 0$ for every ϵ .

Numerical exercise — see the companion code

$w \sim U[0, 10]$, $\epsilon \sim U[0, 4]$ (40 points), $c = 1$, $\phi = .5$, $\beta = .95$. $\bar{w}(\epsilon) = 7.5283, 7.5758, 7.6258$ at $\epsilon = .05, 1.95, 3.95$, slope $0.025 = (1-\beta)(1-\phi)$; after a +1 shift of ϵ : $7.6190, 7.6665, 7.7165$. VFI on the *original* two-dimensional problem gives the same $\bar{w}(\epsilon)$.

Exercise 6.13 Search and Asset Accumulation

Question

(pp. 212–213) A previously unemployed worker gets an offer $w \sim F$; a previously employed worker gets none but may quit (collect z this period, draw next period). Preferences $E \sum \beta^t [u(c_t) + v(l_t)]$, $l_t \in \{0, 1\}$; $a_{t+1} \leq R_{t+1}(a_t + w_t - c_t)$ employed, $a_{t+1} \leq R_{t+1}(a_t + z - c_t)$ unemployed, R i.i.d. with c.d.f. H , observed before $t+1$ choices; $a_t \geq 0$. Write a Bellman equation.

Solution:

State. Assets a (after the return has been realised) and the wage the worker could work at this period: the offer in hand if previously unemployed, the current job's wage if previously employed.

Two functions that coincide. Let $V^u(a, w)$ be the value of a previously unemployed worker holding offer w and $V^e(a, w)$ that of a previously employed worker in a job paying w . Both choose between working at w this period (and next period being "previously

employed at w ") and not working (receiving z , and next period being "previously unemployed" with a fresh draw). Their choice sets and payoffs are identical, so $V^u = V^e \equiv V$: rejecting an offer and quitting a job are the same action. With savings $s = a + y - c \geq 0$ (the no-borrowing constraint $a_{t+1} \geq 0$, given $R > 0$):

$$V(a, w) = \max \left\{ \begin{array}{l} \max_{0 \leq c \leq a+w} u(c) + v(0) + \beta \int V(R'(a + w - c), w) dH(R'), \\ \max_{0 \leq c \leq a+z} u(c) + v(1) + \beta \iint V(R'(a + z - c), w') dF(w') dH(R') \end{array} \right\}$$

u bounded and $\beta < 1$ make the right side a contraction on bounded functions, so V is unique. For each a the worker works iff $w \geq \bar{w}(a)$ (when the accept set is an interval); a previously employed worker with wage w quits in any period in which his assets have risen enough that $\bar{w}(a_t) > w$.

Numerical exercise — see the companion code

Illustration: $u(c) = 1 - e^{-c}$, $v(0) = 0$, $v(1) = .3$, $z = .3$, w uniform on $[0, 2]$ (15 points), $R \in \{.98, 1.06\}$ equally likely, $\beta = .95$, $a \in [0, 15]$ (120 points). VFI converges in 446 iterations; exact evaluation of the VFI policy (a linear system) reproduces V to 10^{-6} . The accept set is an upper interval in w for every a , with $\bar{w}(0) = 1.27$, $\bar{w}(5) = 1.93$, and no wage accepted once $a \geq 7.56$: with bounded utility, richer workers buy leisure, so accumulating assets eventually triggers a quit.

Finite Horizons, Mean-Preserving Spreads and Search for a Price

Exercise 6.9 Search with a Finite Horizon

Question

(p. 210) A worker lives two periods; if unemployed she draws a lifetime-work offer $w \sim F$ each period; objective $E\{y_1 + \beta y_2\}$ with $y = w$ employed, c unemployed. Show the optimal strategy is a reservation wage each period and that the reservation wage decreases over time.

Solution:

Period 2. Accept gives w , reject gives c : $v_2(w) = \max\{w, c\}$, so $\bar{w}_2 = c$.

Period 1. Accepting gives w in both periods, $(1 + \beta)w$; rejecting gives $c + \beta \int v_2 dF$:

$$v_1(w) = \max \left\{ (1 + \beta)w, c + \beta \int \max\{w', c\} dF(w') \right\}.$$

The accept value is increasing in w and the reject value is constant, so accept iff $w \geq \bar{w}_1$

with

$$\bar{w}_1 = \frac{c + \beta \int \max\{w', c\} dF(w')}{1 + \beta}$$

Ranking. $\int \max\{w', c\} dF \geq c$, so $\bar{w}_1 \geq \frac{c + \beta c}{1 + \beta} = c = \bar{w}_2$, strictly if $\Pr\{w > c\} > 0$. The reservation wage falls because the option value of waiting disappears as the horizon shortens: in period 1 a rejected offer can be replaced by a better one next period, in period 2 it cannot.

Numerical exercise — see the companion code

F uniform on $[0, 1]$, $c = .3$, $\beta = .95$: $\bar{w}_1 = 0.41936 > \bar{w}_2 = 0.3$. A brute-force search over all pairs of grid cutoffs (x_1, x_2) picks the grid points just above $(\bar{w}_1, \bar{w}_2)$.

Exercise 6.10 Finite Horizon and Mean-Preserving Spread

Question

(pp. 210–211) Horizon $t = 0, \dots, T$, objective $E \sum_{t=0}^T \beta^t y_t$, $y_t = w$ once employed (forever), zero otherwise; offers i.i.d. from F_i , $i = 1, 2$, where F_1 is an MPS of F_2 and both are bounded by B . (a) Show that the reservation wages coincide at $t = T$ and $t = T - 1$. (b) Argue that for $t \leq T - 2$ the reservation wage under F_1 is higher. (c) With compensation c , prove that (a) fails: under F_1 it is higher at $t = T - 1$.

Solution:

Set-up. Accepting at t pays $A_t w$ with $A_t \equiv \sum_{s=0}^{T-t} \beta^s$. Let Q_t be the value of rejecting at t . Then $V_t(w) = \max\{A_t w, Q_t\}$, the policy is a cutoff $\bar{w}_t = Q_t/A_t$, and

$$Q_T = c, \quad Q_t = c + \beta \int \max\{A_{t+1} w', Q_{t+1}\} dF(w') \quad (t < T),$$

with $c = 0$ in (a)–(b).

Lemma (convex functions and MPS). If F_1 is an MPS of F_2 and φ is convex, then $\int \varphi dF_1 \geq \int \varphi dF_2$. Proof: integrating by parts, $\int_0^B \varphi dF = \varphi(B) - \int_0^B \varphi'(w) F(w) dw$, so with $D(y) \equiv \int_0^y [F_1 - F_2] dw$,

$$\int \varphi dF_1 - \int \varphi dF_2 = - \int_0^B \varphi'(w) dD(w) = -[\varphi' D]_0^B + \int_0^B D(w) d\varphi'(w) = \int_0^B D d\varphi' \geq 0,$$

because $D(0) = 0$, $D(B) = 0$ by (i) (equal means, since $Ew = B - \int_0^B F$), $D \geq 0$ by (iii), and φ' is nondecreasing. For $\varphi(w) = \max\{Aw, Q\}$, φ' jumps by A at the kink Q/A , so the difference is exactly $AD(Q/A)$.

(a) $\bar{w}_T = Q_T/A_T = 0$ under both. At $T - 1$, $Q_{T-1} = \beta \int \max\{w', 0\} dF = \beta Ew$: the continuation is *linear* in w , only the mean matters, and the means are equal. So $\bar{w}_{T-1} = \beta Ew/(1 + \beta)$ for both distributions.

(b) Induction on $Q_t^1 \geq Q_t^2$, true with equality at $T - 1$. If $Q_{t+1}^1 \geq Q_{t+1}^2$,

$$\begin{aligned} Q_t^1 &= \beta \int \max\{A_{t+1}w, Q_{t+1}^1\} dF_1 \geq \beta \int \max\{A_{t+1}w, Q_{t+1}^2\} dF_1 \\ &\geq \beta \int \max\{A_{t+1}w, Q_{t+1}^2\} dF_2 = Q_t^2, \end{aligned}$$

the second step by the lemma. At $t = T - 2$ the second inequality equals $\beta A_{T-1}D(\bar{w}_{T-1}) > 0$ whenever the spread moves mass across $\bar{w}_{T-1}$ (i.e. $D(\bar{w}_{T-1}) > 0$, the generic case), and the strict gap then propagates backwards through the first inequality. Since A_t is common, $\bar{w}_t^1 > \bar{w}_t^2$ for $t \leq T - 2$. From $T - 2$ backwards the continuation contains an *option* (the convex kink), and a riskier distribution makes options more valuable.

(c) Now $V_T(w) = \max\{w, c\}$ already has a kink at c , so $Q_{T-1}^1 - Q_{T-1}^2 = \beta \left[\int \max\{w, c\} dF_1 - \int \max\{w, c\} dF_2 \right] = \beta D(c) > 0$ when the spread moves mass across c . Hence $\bar{w}_{T-1}^1 = Q_{T-1}^1/(1 + \beta) > \bar{w}_{T-1}^2$, while $\bar{w}_T = c$ for both.

Numerical exercise — see the companion code

F_2 triangular on $[0, 1]$, F_1 uniform on $[0, 1]$ (400-point grids; (i) and (iii) verified), $T = 6$, $\beta = .95$. Reservation wages for $t = 0, \dots, 6$:

t	0	1	2	3	4	5	6
$c = 0, F_1$	.5170	.4894	.4545	.4084	.3440	.2436	0
$c = 0, F_2$	.4662	.4453	.4186	.3829	.3310	.2436	0
$c = .2, F_1$	.5792	.5559	.5267	.4885	.4360	.3559	.2
$c = .2, F_2$	.5215	.5040	.4821	.4535	.4132	.3488	.2

Backward induction on the value functions $V_t(w)$ gives identical numbers.

Exercise 6.20 Offer Distributions with Atoms

Question

(pp. 217–218) A type- α worker, $\alpha \in [0, .5]$, draws from $F_\alpha(w) = \alpha$ on $[0, \alpha B]$, w/B on $(\alpha B, B - \alpha B)$, $1 - \alpha$ on $[B - \alpha B, B)$, 1 at B ; compensation $0 < c < B$; no recall, no quits. (a) Bellman equation; prove the reservation wage is time-invariant. (b) Which of $\alpha = 0$ and $\alpha = .3$ has the higher reservation wage? (c) Which type finds a job more quickly?

Solution:

(a) $v(w) = \max\{w/(1 - \beta), c + \beta \int v dF_\alpha\}$. The environment is stationary, the reject value is a constant Q_α and the accept value rises with w : accept iff $w \geq \bar{w}_\alpha = (1 - \beta)Q_\alpha$ in every period. By (6.3.5),

$$\bar{w}_\alpha - c = \beta(E_\alpha w - c) + \beta g_\alpha(\bar{w}_\alpha), \quad g_\alpha(x) = \int_0^x F_\alpha(w) dw.$$

(b) Same mean. $E_\alpha w = \int_0^B [1 - F_\alpha] dw = B - g_\alpha(B)$ and

$$g_\alpha(B) = \alpha \cdot \alpha B + \frac{(B - \alpha B)^2 - (\alpha B)^2}{2B} + (1 - \alpha)\alpha B = \alpha B + \frac{(1 - 2\alpha)B}{2} = \frac{B}{2},$$

so $E_\alpha w = B/2$ for every α (property (i)). *Spread.* $D(y) = g_{.3}(y) - g_0(y)$ equals $.3y - y^2/(2B) > 0$ on $(0, .3B]$, the constant $.3^2 B/2 > 0$ on the middle interval, and falls linearly to $D(B) = 0$ on the top interval. So $D > 0$ on $(0, B)$: $F_{.3}$ is an MPS of F_0 (property (iii)), and $g_{.3} > g_0$ on $(0, B)$. The right side of the reservation equation shifts up while the left side is unchanged, so

$$\boxed{\bar{w}_{.3} > \bar{w}_0} \quad (\text{for every } 0 < c < B).$$

(c) The exit probability is $\lambda_\alpha = \Pr\{w \geq \bar{w}_\alpha\} = 1 - F_\alpha(\bar{w}_\alpha^-)$. For $\alpha = 0$, $\lambda_0 = 1 - \bar{w}_0/B$. For $\alpha = .3$, $\lambda_{.3} = 1 - \bar{w}_{.3}/B$ if $\bar{w}_{.3}$ is in the middle interval, but $\lambda_{.3} = .3$ if $\bar{w}_{.3} \in (.7B, B)$ (only the atom at B is acceptable).

- If $\bar{w}_0 < .7B$: either $\bar{w}_{.3}$ is in the middle interval and $\lambda_{.3} = 1 - \bar{w}_{.3}/B < 1 - \bar{w}_0/B$; or $\bar{w}_{.3} \geq .7B$ and $\lambda_{.3} = .3 < 1 - \bar{w}_0/B$; or $\bar{w}_{.3} \leq .3B$, $\lambda_{.3} = .7 < 1 - \bar{w}_0/B$. The *less risky* type finds a job faster.
- If $\bar{w}_0 > .7B$: then $\bar{w}_{.3} > .7B$ too, $\lambda_{.3} = .3 > 1 - \bar{w}_0/B = \lambda_0$. The *riskier* type finds a job faster, because the atom at B guarantees a 30% chance each period of an acceptable offer.

With F_0 uniform, $\bar{w}_0 - c = \frac{\beta}{1-\beta} \frac{(B-\bar{w}_0)^2}{2B}$, so $\bar{w}_0 = .7B$ at $c^* = .7B - \frac{\beta}{1-\beta} \frac{(.3B)^2}{2B}$. So the answer is: type 0 exits faster if $c < c^*$, type .3 if $c > c^*$. The pure option argument (higher $\bar{w}$ means longer spells) fails because the spread also moves mass to the top of the support.

Numerical exercise — see the companion code

$B = 1, \beta = .9$: $c^* = 0.295$. $c = .1$: $\bar{w}_0 = .6503 < \bar{w}_{.3} = .7568$, $\lambda_0 = .3497 > \lambda_{.3} = .3$. $c = .5$: $\bar{w}_0 = .7597 < \bar{w}_{.3} = .8649$, $\lambda_0 = .2403 < \lambda_{.3} = .3$. The closed-form g_α matches VFI on a discretised F_α with atoms (to 10^{-6}); simulated mean durations match $1/\lambda_\alpha$.

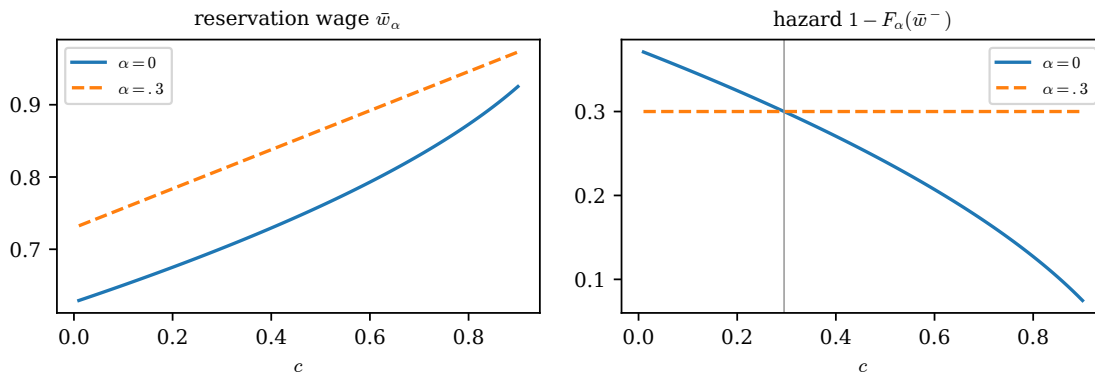


Figure 1: Exercise 6.20. Left: $\bar{w}_{.3} > \bar{w}_0$ for every c . Right: the exit hazards cross at $c^* = 0.295$ (vertical line), where $\bar{w}_0 = .7$ and type .3 starts to accept only the atom at B .

Exercise 6.21 Searching for the Lowest Price

Question

(p. 218) At cost $c > 0$ per draw a buyer draws a price offer $p \sim F$, $F(\underline{B}) = 0$, $F(\bar{B}) = 1$; all search is within one period (no discounting); the buyer minimises price plus search costs. (a) Optimal strategy. (b) Expected price net of all search costs.

Solution:

(a) Let V be the expected total outlay (price plus costs still to be paid) of a buyer about to pay for a draw and behaving optimally. After drawing p he either buys (pays p) or starts over (pays V in expectation):

$$V = c + \int \min\{p, V\} dF(p).$$

So he buys iff $p \leq \bar{p} \equiv V$: a *reservation price*. Rearranging, $c = \int [V - \min\{p, V\}] dF = \int_{\underline{B}}^{\bar{p}} (V - p) dF(p)$, and integrating by parts,

$$c = \int_{\underline{B}}^{\bar{p}} F(p) dp$$

The right side is 0 at $\bar{p} = \underline{B}$ and strictly increasing, so $\bar{p}$ is unique. If $c \geq \int_{\underline{B}}^{\bar{B}} F = \bar{B} - E p$, the equation has no root below $\bar{B}$: the buyer takes the first draw and $\bar{p} = c + E p$ (the formula $\int_{\underline{B}}^{\bar{p}} F = \bar{p} - E p$ for $\bar{p} \geq \bar{B}$ gives the same number).

(b) The expected outlay is $V = \bar{p}$ itself. Decomposing: the number of draws is geometric with success probability $F(\bar{p})$, so expected costs are $c/F(\bar{p})$, and the price paid has mean $E[p \mid p \leq \bar{p}]$; indeed $E[p \mid p \leq \bar{p}] + c/F(\bar{p}) = \bar{p}$, because $c = \int_{\underline{B}}^{\bar{p}} (\bar{p} - p) dF = F(\bar{p})\{\bar{p} - E[p \mid p \leq \bar{p}]\}$. (We read "purchase price net of all search costs" as price *plus* search costs, the buyer's total outlay.)

Numerical exercise — see the companion code

F uniform on $[1, 2]$: $\bar{p} = 1 + \sqrt{2c}$ for $c \leq \frac{1}{2}$, $\bar{p} = c + 1.5$ otherwise. $c = .02$: 1.2 (simulated outlay 1.1999, 5.0 draws); $c = .1$: 1.4472 (1.4482); $c = .6$: 2.1 (one draw). Minimising $E[p \mid p \leq x] + c/F(x)$ over x gives the same $\bar{p}$.

Time-Varying Reservation Wages: Temporary Benefits and Seasons

Exercise 6.14 Temporary Unemployment Compensation

Question

(pp. 213–214) One offer per period, $F(0) = 0$, $F(B) = 1$; compensation $\gamma > 0$ only in the *first* period of unemployment. (a) Bellman equations; the policy is a time-varying reservation wage. (b) How does it vary with duration? (c) How does the hazard vary? Then compensation is also paid after a quit (first period of a spell only; one period of work requalifies); an employed worker decides at the start of a period whether to quit and, if so, draws an offer at once. (d) Bellman equations for $v^e(w)$, $v_1^u(w')$, $v_+^u(w')$. (e) Characterise $\bar{w}^e$, $\bar{w}_1^u$, $\bar{w}_+^u$ and relate them to γ .

Solution:

(a) Let v_1 be the value in the first period of a spell (offer in hand, eligible for γ) and v_+ afterwards:

$$v_1(w) = \max \left\{ \frac{w}{1-\beta}, \gamma + \beta \int v_+ dF \right\}, \quad v_+(w) = \max \left\{ \frac{w}{1-\beta}, \beta \int v_+ dF \right\}$$

Both reject values are constants, so each period there is a reservation wage: $\bar{w}_1 = (1-\beta)(\gamma + \beta E v_+)$ in the first period, $\bar{w}_+ = (1-\beta)\beta E v_+$ afterwards.

(b) v_+ is McCall with $c = 0$: $\bar{w}_+ = \frac{\beta}{1-\beta} \int_{\bar{w}_+}^B (w' - \bar{w}_+) dF(w')$. And

$$\bar{w}_1 = \bar{w}_+ + (1-\beta)\gamma.$$

The reservation wage drops by $(1-\beta)\gamma$ after the first period and is constant thereafter.

(c) The hazard is $1 - F(\bar{w}_1)$ in the first period and $1 - F(\bar{w}_+) \geq 1 - F(\bar{w}_1)$ from the second period on: it rises (weakly) once, then stays flat.

(d) Accepting w pays w now and makes the worker "previously employed at w " next period:

$$\begin{aligned} v^e(w) &= \max \left\{ w + \beta v^e(w), \int v_1^u(w') dF(w') \right\}, \\ v_1^u(w') &= \max \left\{ w' + \beta v^e(w'), \gamma + \beta \int v_+^u dF \right\}, \\ v_+^u(w') &= \max \left\{ w' + \beta v^e(w'), \beta \int v_+^u dF \right\}. \end{aligned}$$

(e) Write $E_1 = \int v_1^u dF$, $E_+ = \int v_+^u dF$ and $A(w) = w + \beta v^e(w)$ (value of working at w this period).

1. *Quits.* By stationarity a worker either stays forever or quits now: $v^e(w) = \max\{w/(1-\beta), E_1\}$, so $\bar{w}^e = (1-\beta)E_1$, and $A(w) = w/(1-\beta)$ for $w \geq \bar{w}^e$, $A(w) = w + \beta E_1$ for $w < \bar{w}^e$.
2. *Ineligible unemployed.* $v_1^u \geq v_+^u$ pointwise ($\gamma > 0$), so $E_1 \geq E_+$. For every $w \geq 0$, $A(w) \geq w + \beta E_1 \geq \beta E_+$: every offer is accepted, $\bar{w}_+^u = 0$. Taking any job requalifies the worker, and he can quit next period with the benefit in hand.

3. *Eligible unemployed.* $E_1 \geq \gamma + \beta E_+$ (as $v_1^u \geq \gamma + \beta E_+$), so at $w = \bar{w}^e$, $A = E_1 \geq \gamma + \beta E_+$: hence $\bar{w}_1^u \leq \bar{w}^e$ and on that range $A = w + \beta E_1$, giving $\bar{w}_1^u = \gamma - \beta(E_1 - E_+)$. Since all offers are accepted by the ineligible, $E_+ = \int A dF$ and $E_1 - E_+ = \int (\gamma + \beta E_+ - A)^+ dF = \int_0^{\bar{w}_1^u} (\bar{w}_1^u - w) dF = \int_0^{\bar{w}_1^u} F(w) dw$. So

$$\bar{w}_1^u + \beta \int_0^{\bar{w}_1^u} F(w) dw = \gamma$$

The left side rises strictly from 0, so $0 < \bar{w}_1^u \leq \gamma$, increasing in γ ; it depends only on γ, β, F .

4. *The parameter-dependent one.* Substituting $E_1 = \bar{w}^e / (1 - \beta)$ in $E_1 = E_+ + \int_0^{\bar{w}_1^u} F$ with $E_+ = \int_{w < \bar{w}^e} (w + \beta E_1) dF + \int_{w \geq \bar{w}^e} \frac{w}{1 - \beta} dF$, and using $E \max\{w, x\} = Ew + \int_0^x F$:

$$\bar{w}^e = Ew + \beta \int_0^{\bar{w}^e} F(w) dw + (1 - \beta) \int_0^{\bar{w}_1^u} F(w) dw$$

So $\bar{w}^e$ exceeds the mean wage and depends on the whole of F , on β , and on γ only through the small last term. Workers holding $w < \bar{w}^e$ cycle: work one period, quit, collect γ if the new draw is below $\bar{w}_1^u$, accept anything the next period, and so on.

Numerical exercise — see the companion code

F uniform on $[0, 1]$, $\beta = .95$. (a)–(c): $\bar{w}_+ = .7239$ for all γ ; $\bar{w}_1 = .7264, .7289, .7339$ for $\gamma = .05, .1, .2$; hazards .270 then .275 at $\gamma = .1$ (simulated hazards by duration agree to 4×10^{-3}). (d)–(e): $\bar{w}_+^u = 0$; $\bar{w}_1^u = .0489, .0957, .1839$; $\bar{w}^e = .8175, .8183, .8211$. VFI on the three value functions reproduces all three cutoffs and the identity $E_1 - E_+ = \int_0^{\bar{w}_1^u} F$.

Exercise 6.15 Seasons, I

Question

(p. 214) In odd periods the unemployed worker gets one offer from F , in even periods two; jobs last forever. (a) Bellman equations. (b) Form of the optimal policy.

Solution:

(a) Let $v_o(w)$, $v_e(w)$ be the values with best offer w in hand in an odd (one-offer) and an even (two-offer) period. Tomorrow is of the other type:

$$v_o(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \int v_e dF^2 \right\}, \quad v_e(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \int v_o dF \right\}$$

(b) Two reservation wages, $\bar{w}_o = (1 - \beta)Q_o$ and $\bar{w}_e = (1 - \beta)Q_e$, with $Q_o = c +$

$\beta \int \max\{\frac{w}{1-\beta}, Q_e\} dF^2$, $Q_e = c + \beta \int \max\{\frac{w}{1-\beta}, Q_o\} dF$. *Claim:* $\bar{w}_o > \bar{w}_e$. Suppose $Q_o \leq Q_e$. Since $\varphi_Q(w) = \max\{w/(1-\beta), Q\}$ is nondecreasing in w and in Q , and F^2 first-order dominates F ,

$$Q_o = c + \beta \int \varphi_{Q_e} dF^2 \geq c + \beta \int \varphi_{Q_o} dF^2 \geq c + \beta \int \varphi_{Q_o} dF = Q_e,$$

so $Q_o = Q_e$ and both inequalities bind; but the second is strict whenever $(1-\beta)Q_o < B$ (then $\int \varphi d(F^2 - F) = \frac{1}{1-\beta} \int_{\bar{w}}^B F(1-F) dw > 0$). Contradiction, so $Q_o > Q_e$. The worker is choosier in a one-offer period, because tomorrow brings two offers; the policy alternates between a high and a low cutoff. Both lie between the stationary one-offer and two-offer reservation wages.

Numerical exercise — see the companion code

F uniform on $[0, 1]$, $c = .2$, $\beta = .95$: $\bar{w}_o = .79794 > \bar{w}_e = .78744$; stationary benchmarks .75770 (one offer always) and .81434 (two always). Two-equation root = VFI on both functions.

Exercise 6.16 Seasons, II

Question

(pp. 214–215) One offer per period for the unemployed, no recall; the wage is fixed while in the job. At the beginning of each odd period an employed worker is fired with probability $\pi \in (0, 1)$ and immediately draws a new offer; no firing in even periods. (a) Bellman equation. (b) Form of the optimal policy.

Solution:

(a) Let $J_o(w)$, $J_e(w)$ be the values of working at w during an odd/even period (after the firing draw), and v_o , v_e the values of an unemployed worker with offer w in hand; $E_x = \int v_x dF$:

$$\begin{aligned} J_o(w) &= w + \beta J_e(w), & J_e(w) &= w + \beta[(1-\pi)J_o(w) + \pi E_o], \\ v_o(w) &= \max\{J_o(w), c + \beta E_e\}, & v_e(w) &= \max\{J_e(w), c + \beta E_o\}. \end{aligned}$$

(A fired worker immediately holds a fresh offer in an odd period, worth E_o in expectation.)

(b) Solving the two linear equations, with $d \equiv 1 - \beta^2(1 - \pi)$:

$$J_e(w) = \frac{[1 + \beta(1 - \pi)]w + \beta\pi E_o}{d}, \quad J_o(w) = w + \beta J_e(w) = \frac{(1 + \beta)w + \beta^2\pi E_o}{d},$$

both strictly increasing, with slopes $s_e = \frac{1+\beta(1-\pi)}{d}$, $s_o = \frac{1+\beta}{d}$. The reject values are constants, so the policy is **two reservation wages**: accept iff $w \geq \bar{w}_o$ in odd periods and iff $w \geq \bar{w}_e$ in even periods, where $J_o(\bar{w}_o) = c + \beta E_e$ and $J_e(\bar{w}_e) = c + \beta E_o$.

Which is higher? By the reduction of the overview, $E_o = c + \beta E_e + s_o G_o$ and $E_e = c + \beta E_o + s_e G_e$ with $G_x \equiv G_F(\bar{w}_x)$. Hence $(1 - \beta^2)E_o = (1 + \beta)c + \beta s_e G_e + s_o G_o$.

- From $J_e(\bar{w}_e) = c + \beta E_o$: $[1 + \beta(1 - \pi)]\bar{w}_e = dc + \beta(d - \pi)E_o$, and $d - \pi = (1 - \pi)(1 - \beta^2)$. Substituting E_o and using $d + \beta(1 - \pi)(1 + \beta) = 1 + \beta(1 - \pi)$:

$$\bar{w}_e = c + \frac{\beta(1 - \pi) [\beta s_e G_e + s_o G_o]}{1 + \beta(1 - \pi)}.$$

- From $J_o(\bar{w}_o) = c + \beta E_e = c(1 + \beta) + \beta^2 E_o + \beta s_e G_e$: $(1 + \beta)\bar{w}_o = (1 + \beta)dc + \beta^2(d - \pi)E_o + \beta d s_e G_e$; eliminating E_o with the first bullet's $\beta(d - \pi)E_o = [1 + \beta(1 - \pi)]\bar{w}_e - dc$ and $d s_e = 1 + \beta(1 - \pi)$:

$$(1 + \beta)\bar{w}_o = dc + \beta[1 + \beta(1 - \pi)](\bar{w}_e + G_e).$$

- Subtracting $(1 + \beta)\bar{w}_e$ and using $(1 + \beta) - \beta[1 + \beta(1 - \pi)] = d$, then the first bullet for $c - \bar{w}_e$ (with $d s_e = 1 + \beta(1 - \pi)$, $d s_o = 1 + \beta$):

$$\bar{w}_o - \bar{w}_e = \frac{\beta}{1 + \beta} [G_F(\bar{w}_e) - \kappa G_F(\bar{w}_o)], \quad \kappa = \frac{(1 - \pi)(1 + \beta)}{1 + \beta(1 - \pi)} \in [0, 1]$$

Fix $\bar{w}_e$ and let $\psi(y) = y - \bar{w}_e - \frac{\beta}{1 + \beta} [G_F(\bar{w}_e) - \kappa G_F(y)]$. Then $\psi'(y) = 1 - \frac{\beta\kappa}{1 + \beta} [1 - F(y)] > 0$, $\psi(\bar{w}_o) = 0$ and $\psi(\bar{w}_e) = -\frac{\beta(1 - \kappa)}{1 + \beta} G_F(\bar{w}_e) < 0$ for $\pi > 0$, $\bar{w}_e < B$. Hence $\bar{w}_o > \bar{w}_e$ (equal when $\pi = 0$). Intuition: firing hands the worker a fresh draw with no unemployment spell, which is valuable to someone holding a *marginal* wage. A job accepted in an even period faces its first firing draw after one period; one accepted in an odd period locks the worker in for two. So the marginal wage is more attractive in even periods. For the same reason both cutoffs fall as π rises.

Numerical exercise — see the companion code

F uniform on $[0, 1]$, $\beta = .95$. $\pi = .3$, $c = .2$: $\bar{w}_o = .57077$, $\bar{w}_e = .56051$. Over $\pi \in \{.1, .5, .9\} \times c \in \{0, .4, .8\}$ the gap ranges from $+.0002$ ($\pi = .1$, $c = .8$) to $+.178$ ($\pi = .9$, $c = 0$), and the boxed identity holds to 10^{-8} . VFI on the four functions agrees with the closed form of J plus a two-equation root.

Quits, Promotions, Human Capital and Markov Wages

Question

(pp. 218–219) One offer per period from F for the unemployed. At the start of each period a worker employed at w has his job reclassified with probability $\alpha \in (0, 1)$: he gets a new wage $w' \sim F$ and may work at w' until reclassified again, or quit (get c now, draw next period). (a) Bellman equation for an unemployed worker. (b) Decision rule of a previously unemployed worker. (c) Quit rule of a previously employed worker. (d) How to compute the probability that a previously employed worker quits.

Solution:

(a) Let $W(w)$ be the value of working at w this period. Next period, with probability α the job is reclassified and the worker is in exactly the position of an unemployed worker holding offer w' :

$$v(w) = \max\{W(w), c + \beta Ev\}, \quad W(w) = w + \beta[(1 - \alpha)W(w) + \alpha Ev]$$

$$Ev = \int v dF$$

so $W(w) = \frac{w + \beta\alpha Ev}{1 - \beta(1 - \alpha)}$, linear and increasing.

(b) Accept iff $W(w) \geq c + \beta Ev$, i.e. $w \geq \bar{w}$. Let $D = 1 - \beta(1 - \alpha)$ and $Q = c + \beta Ev$. Then $Ev = Q + \frac{1}{D}G_F(\bar{w})$, so $(1 - \beta)Ev = c + G_F(\bar{w})/D$. From $W(\bar{w}) = Q$: $\bar{w} = DQ - \beta\alpha Ev = Dc + \beta(D - \alpha)Ev$ and $D - \alpha = (1 - \alpha)(1 - \beta)$, so $\bar{w} = Dc + \beta(1 - \alpha)[c + G_F(\bar{w})/D]$, and with $D + \beta(1 - \alpha) = 1$:

$$\bar{w} - c = \frac{\beta(1 - \alpha)}{1 - \beta(1 - \alpha)} \int_{\bar{w}}^B (w - \bar{w}) dF(w)$$

McCall with the effective discount factor $\beta(1 - \alpha)$: good jobs are less valuable because they may be reclassified, so the worker is less choosy.

(c) A reclassified worker faces the same comparison $W(w')$ vs. $c + \beta Ev$: he keeps the new wage iff $w' \geq \bar{w}$ and quits otherwise. A worker who is not reclassified keeps $w \geq \bar{w}$ (quitting would give $Q \leq W(w)$).

(d) Each period, $\Pr\{\text{quit} \mid \text{employed}\} = \alpha F(\bar{w})$, with $\bar{w}$ from (b). The number of periods until a quit is geometric with mean $1/[\alpha F(\bar{w})]$.

Numerical exercise — see the companion code

F uniform on $[0, 1]$, $c = .3$, $\beta = .95$, $\alpha = .2$: $\bar{w} = .57970$, quit probability $\alpha F(\bar{w}) = .11600$; a simulation of 50 000 workers for 200 periods, with decisions read off the VFI arrays, gives .11607.

Exercise 6.23 A Career Ladder

Question

(p. 219) One offer per period from F for the unemployed. At the start of each period a worker employed at w is promoted with probability α to γw , $\gamma > 1$, until the next promotion. (a) Bellman equation for a previously employed worker. (b) For a previously unemployed worker. (c) Decision rule of an unemployed worker. (d) Decision rule of a previously employed worker.

Solution:

(a) $W(w) = w + \beta[(1 - \alpha)W(w) + \alpha W(\gamma w)]$. Guess $W(w) = kw$: $k = 1 + \beta(1 - \alpha)k + \beta\alpha\gamma k$, so

$$W(w) = \frac{w}{\kappa}, \quad \kappa \equiv 1 - \beta(1 - \alpha + \alpha\gamma),$$

which requires $\beta(1 - \alpha + \alpha\gamma) < 1$ (otherwise the value is infinite). Check: $E[w_t] = w(1 - \alpha + \alpha\gamma)^t$, so $\sum_t \beta^t E[w_t] = w/\kappa$.

(b) $v(w) = \max\{w/\kappa, c + \beta \int v dF\}$.

(c) Accept iff $w \geq \bar{w}$. With $Q = c + \beta Ev$ and $Ev = Q + G_F(\bar{w})/\kappa$, $\bar{w} = \kappa Q$: $\frac{\bar{w}}{\kappa} = c + \frac{\beta\bar{w}}{\kappa} + \frac{\beta G_F(\bar{w})}{\kappa}$, i.e.

$$(1 - \beta)\bar{w} = \kappa c + \beta \int_{\bar{w}}^B (w - \bar{w}) dF(w)$$

This is Exercise 6.8 with ϕ replaced by the expected growth factor $1 - \alpha + \alpha\gamma$: promotions lower the reservation wage when $c > 0$ ($d\bar{w}/d\gamma = -\beta\alpha c/[1 - \beta F(\bar{w})] < 0$).

(d) A previously employed worker stays and takes every promotion: wages only rise, so $W(\gamma^j w) \geq W(w) \geq Q$ for $w \geq \bar{w}$ and quitting (were it allowed) would never pay.

Numerical exercise — see the companion code

F uniform on $[0, 1]$, $c = .3$, $\beta = .95$, $\alpha = .1$, $\gamma = 1.2$: $1/\kappa = 32.258$ (Monte Carlo of promotion paths: 32.30); $\bar{w} = .75522 < .77613$ (no promotions). VFI agrees to 10^{-9} .

Exercise 6.24 Human Capital

Question

(pp. 219–220) Offers pay wh , $w \sim F$; w is kept during the job, h follows a Markov chain H^e when employed (tends to rise) and H^u when unemployed (tends to fall). At the start of a period employed workers observe the new h and may quit, keeping h and drawing a new w at once (accept it, or be unemployed at least one period). (a) Bellman equation(s). (b) Describe the optimal rule; do employed workers ever quit? (c) An algorithm.

Solution:

(a) Let $S(w, h)$ be the value of working at w this period with human capital h ; $V^e(w, h)$ the value of a worker employed last period at w who has just observed h ; $U(h)$ the value of an unemployed worker about to draw w :

$$\begin{aligned} S(w, h) &= wh + \beta \sum_j H^e(h, j) V^e(w, \bar{h}_j), & V^e(w, h) &= \max\{S(w, h), U(h)\}, \\ U(h) &= \int \max\left\{S(w, h), c + \beta \sum_j H^u(h, j) U(\bar{h}_j)\right\} dF(w). \end{aligned}$$

(b) By induction on the iterations, S is nondecreasing in w , so both decisions are cutoffs:

- unemployed: accept iff $w \geq \bar{w}^u(h)$, where $S(\bar{w}^u(h), h) = R^u(h) \equiv c + \beta \sum_j H^u(h, j) U(\bar{h}_j)$;
- employed: stay iff $w \geq \bar{w}^e(h)$, where $S(\bar{w}^e(h), h) = U(h)$.

Since $U(h) = \int \max\{S, R^u(h)\} dF \geq R^u(h)$, $\bar{w}^e(h) \geq \bar{w}^u(h)$, strictly when the draw has option value. So employed workers *do* quit: anyone whose w lies below $\bar{w}^e(h)$ at his current h — in particular someone who accepted $w \in [\bar{w}^u, \bar{w}^e)$ quits as soon as next period. Accepting such a job is still worthwhile: employment makes h drift up (a rejecting worker's h drifts down), it pays wh now, and quitting later gives an *immediate* redraw, whereas rejecting costs a period of search. Quitting is this model's form of on-the-job search. Rising h also changes both cutoffs, so quits can follow human capital shocks.

(c) (1) Grids for w and the n values $\bar{h}$. (2) Iterate the three equations as a map $(V^e, U) \mapsto (V^e, U)$; it is a contraction with modulus β (bounded payoffs, discounted expectations), so it converges from any start. (3) Read off $\bar{w}^u(h)$, $\bar{w}^e(h)$. (4) Faster: Howard's algorithm — fix the cutoff rules, solve the linear system for (S, U) , improve, repeat.

Numerical exercise — see the companion code

$\bar{h} \in \{1, 1.25, \dots, 2\}$; H^e : up .4, stay .5, down .1; H^u : up .1, stay .5, down .4 (reflecting); w uniform on $[0, 1]$ (100 points), $c = .5$, $\beta = .95$. VFI (507 iterations) and Howard (6 improvements) give identical rules. Acceptance cutoffs $\bar{w}^u = (.165, .005, .005, .065, .185)$, quit cutoffs $\bar{w}^e = (.845, .835, .825, .825, .815)$ across h : the unemployed take almost any job and then quit into better draws. A simulated panel shows positive quits.

Exercise 6.25 Markov Wages**Question**

(pp. 220–221) Successive offers follow a Markov chain on $\bar{w}_1 < \dots < \bar{w}_n$ with transition matrix P ; jobs are permanent. (a) Bellman equation. (b) Can you prove the reservation-wage property? (c) With $\beta = .95$, $c = 1$, $\bar{w} = (1, \dots, 5)$ and the given P , compute the optimal policy and value. (d) Same with $\beta = .99$. (e) Discuss.

Solution:

$$(a) \quad v_i = \max \left\{ \frac{\bar{w}_i}{1 - \beta}, c + \beta \sum_j P_{ij} v_j \right\}, i = 1, \dots, n.$$

(b) Not in general. The reject value $c + \beta P_i v$ now depends on the current state because today's offer predicts tomorrow's. The accept set is an upper set only if $\bar{w}_i/(1 - \beta) - \beta P_i v$ is increasing in i , which holds, e.g., for i.i.d. offers (identical rows) but can fail when a low wage forecasts high future wages. The example has such a state: from $\bar{w}_3$ the chain jumps to 4 or 5 half the time.

(c)–(d) VFI, cross-checked by enumerating all $2^5 = 32$ accept/reject rules (each evaluated by solving $v = r + \beta P_{\text{rej}} v$): the VFI value dominates all of them and is attained by the rule shown.

state i ($\bar{w}_i$)	1	2	3	4	5
$\beta = .95$: accept?	no	yes	no	no	yes
$\beta = .95$: v_i	35.833	40.000	61.834	80.312	100.000
$\beta = .99$: accept?	no	no	no	no	yes
$\beta = .99$: v_i	224.188	230.460	352.798	466.757	500.000

(e) With $\beta = .95$ the policy is *not* a reservation wage: $\bar{w}_2 = 2$ is accepted while 3 and 4 are rejected. State 2 is persistent and mostly leads back to 1 or 2, so waiting there is expensive; state 3 is a springboard (half the time next period's offer is 4 or 5), and state 4 is one step from 5 (probability .18 per period). Check at state 2: rejecting is worth $1 + .95(.18 \cdot 35.83 + .8 \cdot 40 + .02 \cdot 61.83) = 38.70 < 40$. With $\beta = .99$ the worker is patient enough to hold out for the top wage from every state: $5/(1 - \beta) = 500$ dwarfs the cost of a longer spell, and the policy becomes a (trivial) reservation wage $\bar{w}_5$. Higher patience raises all values, weakly shrinks the accept set and restores monotonicity here.

Search Intensity, Externalities and On-the-Job Investment

Exercise 6.6 Mortensen Externality

Question

(pp. 208–209) Worker and firm jointly draw a match $\theta \sim F$; matched pairs stay together forever, each getting θ per period. Each side receives $n = f(c_1 + c_2)$ offers per period, where c_1, c_2 are the resources the worker and the typical firm devote to search, chosen at the start of the period; both have the same reservation θ . (a) Nash equilibrium: Bellman equations for Q_i , the value of an unmatched type- i agent before choosing c_i . (b) Planner maximising $\lambda \cdot (\text{utility 1}) + (1 - \lambda) \cdot (\text{utility 2})$: Bellman equation for $Q(\lambda)$. (c) Argue that Nash search is not optimal.

Solution:

Timing: costs are paid, the n draws arrive, the best one is accepted or not; an accepted match pays θ from this period on, worth $\theta/(1 - \beta)$; an unmatched agent gets nothing this period. Define

$$M(n, R) \equiv \int \max \left\{ \frac{\theta}{1 - \beta}, R \right\} dF^n(\theta) = R + \frac{1}{1 - \beta} \int_{\bar{\theta}}^B [1 - F(\theta)^n] d\theta, \quad \bar{\theta} = (1 - \beta)R,$$

the expected value of the best of n draws when rejecting is worth R .

(a) Taking c_j as given,

$$Q_i = \max_{c_i \geq 0} \left\{ -c_i + \int \max \left\{ \frac{\theta}{1 - \beta}, \beta Q_i \right\} dF^{f(c_i + c_j)}(\theta) \right\}$$

that is, $Q_i = \max_{c_i} \{-c_i + M(f(c_i + c_j), \beta Q_i)\}$. Interior first-order condition: $1 = f'(c_1 + c_2) M_n(f(c_1 + c_2), \beta Q_i)$, where $M_n = -\frac{1}{1 - \beta} \int_{\bar{\theta}}^B F^n \ln F d\theta > 0$.

(b) Both parties receive the same θ , so the weighted benefit is $\lambda\theta + (1 - \lambda)\theta = \theta$, while the weighted cost is $\lambda c_1 + (1 - \lambda)c_2$:

$$Q(\lambda) = \max_{c_1, c_2 \geq 0} \left\{ -\lambda c_1 - (1 - \lambda)c_2 + \int \max \left\{ \frac{\theta}{1 - \beta}, \beta Q(\lambda) \right\} dF^{f(c_1 + c_2)}(\theta) \right\}$$

Only $C = c_1 + c_2$ matters for the benefit, so the planner loads search on the cheaper party and the first-order condition is $\min\{\lambda, 1 - \lambda\} = f'(C) M_n(f(C), \beta Q(\lambda))$.

(c) In Nash, each party equates its *private* marginal cost, 1, to its private marginal benefit $f'M_n$; but an extra unit of c_i also raises the partner's chance of a good match by exactly the same amount, an external benefit it ignores. The planner equates $f'M_n$ to $\min\{\lambda, 1 - \lambda\} \leq \frac{1}{2} < 1$. With f concave and $M_{nn} = -\frac{1}{1 - \beta} \int F^n (\ln F)^2 d\theta < 0$, the product $f'(C)M_n(f(C), R)$ falls with C , so at a given continuation value R the planner chooses a larger C : **too few resources go to search in the Nash equilibrium.** (The

planner's higher R lowers M_n somewhat; the numerical example confirms the ranking with R endogenous.)

Numerical exercise — see the companion code

$\theta \sim U[0, 1]$, $n = f(C) = 1 + 4\sqrt{C}$ (real n , c.d.f. θ^n), $\beta = .9$. Nash: $c_1 = c_2 = .1111$, $C = .2222$ ($n = 2.886$), $Q_i = 8.0516$. Planner $\lambda = .5$: $C = .5495$ ($n = 3.965$), value per agent $8.1412 > 8.0516$; $\lambda = .3$: $C = .9453$. The closed form for M matches quadrature; the Nash c is a best response by direct maximisation and satisfies the FOC.

Exercise 6.11 Pissarides: Taxation and Variable Search Intensity

Question

(p. 211) With effort e the unemployed worker gets an offer $w \sim F$ with probability $\pi(e)$, $\pi(0) = 0$, $\pi' > 0$, $\pi'' < 0$; a worker who accepts is fired with probability α per period. Flows: $1 - e + z$ when searching and not accepting; $w - e - T(w)$ in the first period of a job; $w - T(w)$ afterwards (taxes T , $w - T(w)$ increasing). (a) Show the policy is a reservation wage independent of e ; display the condition for e . (b) With $T(w) = t(w - a)$, show that a higher a lowers the reservation wage and raises effort. (c) When does a change in t have the opposite effect?

Solution:

(a) Let $n(w) = w - T(w)$, $D = 1 - \beta(1 - \alpha)$. An employed worker's value (a fired worker is unemployed next period, with value U before choosing effort) is

$$W(w) = n(w) + \beta[(1 - \alpha)W(w) + \alpha U] \implies W(w) = \frac{n(w) + \beta\alpha U}{D}.$$

If the worker searches with effort e and draws w : accept gives $W(w) - e$; reject gives $1 - e + z + \beta U$.

$$U = \max_e \left\{ (1 - \pi(e))(1 - e + z + \beta U) + \pi(e) \int \max\{W(w) - e, 1 - e + z + \beta U\} dF(w) \right\}$$

The effort cost e appears in both branches, so the accept/reject comparison is $W(w)$ vs. $1 + z + \beta U$, independent of e ; W is increasing (as n is), so accept iff $w \geq \bar{w}$, $W(\bar{w}) = 1 + z + \beta U$. Pulling $1 - e + z + \beta U$ out,

$$U = 1 + z + \beta U + \max_e \left\{ \pi(e)\Gamma(\bar{w}) - e \right\}, \quad \Gamma(\bar{w}) \equiv \int_{\bar{w}}^B [W(w) - W(\bar{w})] dF,$$

where $\Gamma(\bar{w}) = \frac{1}{D} \int_{\bar{w}}^B [n(w) - n(\bar{w})] dF$. So the effort condition is

$$\pi'(e)\Gamma(\bar{w}) = 1 \quad (\text{interior}).$$

The reservation equation. From $W(\bar{w}) = 1 + z + \beta U$: $n(\bar{w}) = D(1 + z) + \beta(D - \alpha)U$, with $D - \alpha = (1 - \alpha)(1 - \beta)$ and $(1 - \beta)U = 1 + z + \pi(e)\Gamma - e$. Since $D + \beta(1 - \alpha) = 1$,

$$n(\bar{w}) = 1 + z + \beta(1 - \alpha) \max_e \{ \pi(e)\Gamma(\bar{w}) - e \}.$$

(b) With $T = t(w - a)$: $n(w) = (1 - t)w + ta$, $\Gamma = (1 - t)G_F(\bar{w})/D$. Define

$$\begin{aligned} \Phi(\bar{w}; a, t) &= (1 - t)\bar{w} + ta - 1 - z - \beta(1 - \alpha) \mathcal{M}(\bar{w}, t), \\ \mathcal{M} &= \max_e \left\{ \pi(e) \frac{(1 - t)G_F(\bar{w})}{D} - e \right\}. \end{aligned}$$

By the envelope theorem $\mathcal{M}_{\bar{w}} = -\pi(e)(1 - t)[1 - F'(\bar{w})]/D$, so $\Phi_{\bar{w}} = (1 - t)[1 + \beta(1 - \alpha)\pi(e)(1 - F'(\bar{w}))/D] > 0$, and $\Phi_a = t > 0$. Hence

$$\frac{d\bar{w}}{da} = -\frac{t}{\Phi_{\bar{w}}} < 0.$$

Effort solves $\pi'(e) = D/[(1 - t)G_F(\bar{w})]$; $\bar{w} \downarrow \Rightarrow G_F(\bar{w}) \uparrow \Rightarrow \pi'(e) \downarrow \Rightarrow e \uparrow$ ($\pi'' < 0$). Both raise the acceptance probability $\pi(e)[1 - F'(\bar{w})]$. A larger exemption raises the net income of every job by ta without touching the unemployed.

(c) $\mathcal{M}_t = -\pi(e)G_F(\bar{w})/D$, so $\Phi_t = a - \bar{w} + \beta(1 - \alpha)\pi(e)G_F(\bar{w})/D$. Using the reservation equation, $\beta(1 - \alpha)\pi(e)G_F/D = [(1 - t)\bar{w} + ta - 1 - z + \beta(1 - \alpha)e]/(1 - t)$, which gives

$$\Phi_t = \frac{a - (1 + z) + \beta(1 - \alpha)e}{1 - t}, \quad \frac{d\bar{w}}{dt} = -\frac{\Phi_t}{\Phi_{\bar{w}}} > 0 \iff \boxed{a < 1 + z - \beta(1 - \alpha)e}.$$

Under that condition a higher t raises $\bar{w}$; and effort falls, since $\pi'(e) = D/[(1 - t)G_F(\bar{w})]$ rises both because $1 - t$ falls and because $G_F(\bar{w})$ falls. That is the opposite of an increase in a . When the exemption is large, $a > 1 + z - \beta(1 - \alpha)e$, the tax is a net subsidy at the margin (it is negative for $w < a$), and $\bar{w}$ falls with t ; the effect on effort is then ambiguous.

Numerical exercise — see the companion code

$\pi(e) = 1 - e^{-2e}$, F uniform on $[0, 5]$, $\beta = .95$, $\alpha = .1$, $z = .5$. At $a = 1$, $t = .3$: $\bar{w} = 2.8723$, $e = .7375$ (literal VFI on w and e grids: 2.8725, .7375); $d\bar{w}/da = -.146$, $de/da = +.069$. At $a = .2$: threshold .919 $> a$, $d\bar{w}/dt = +.532$, $de/dt = -.979$. At $a = 2.5$: threshold .790 $< a$, $d\bar{w}/dt = -1.080$, $de/dt = -.252$. The analytic $d\bar{w}/dt$ matches the finite difference to 10^{-5} .

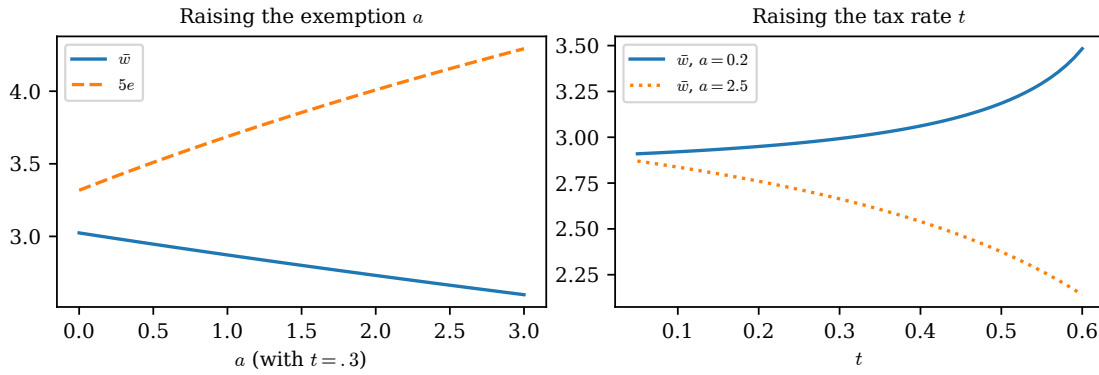


Figure 2: Exercise 6.11. Left: raising the exemption a lowers $\bar{w}$ and raises effort. Right: raising t raises $\bar{w}$ when $a = .2$ (below the threshold) and lowers it when $a = 2.5$.

Exercise 6.18 Jovanovic (1979b)

Question

(p. 216) An employed worker earns $w_t = x_t(1 - \phi_t - s_t)$: x_t job-specific capital, ϕ_t time invested in it, s_t time searching. Searching s_t brings, with probability $\pi(s_t)$, an offer to restart at capital $\mu' \sim F$ next period ($\pi' > 0$, $\pi'' < 0$). On the job $x_{t+1} = G(x_t, \phi_t)$, printed as $g(x_t\phi_t) - \delta x_t$; $x_0 = \mu$. Risk neutral, discount β . (a) Bellman equation. (b) Rule for accepting μ' . (c) With $g(x\phi) = A(x\phi)^\alpha$, $\pi(s) = s^{-5}$, F discrete on n values, compute and display the optimal ϕ , s .

Solution:

Misprint. As printed, $x_{t+1} = g(x_t\phi_t) - \delta x_t$ is negative whenever ϕ_t is small (e.g. $\phi_t = 0$ gives $-\delta x_t$), which is impossible for a capital stock and inconsistent with calling δ a depreciation rate. We use

$$x_{t+1} = G(x_t, \phi_t) = (1 - \delta)x_t + g(x_t\phi_t).$$

(a) Let $x' = G(x, \phi)$:

$$V(x) = \max_{\phi, s \geq 0, \phi + s < 1} \left\{ x(1 - \phi - s) + \beta \left[(1 - \pi(s))V(x') + \pi(s) \int \max\{V(x'), V(\mu')\} dF(\mu') \right] \right\}$$

(b) V is increasing: more capital raises this period's earnings for any (ϕ, s) and, since $G_x = (1 - \delta) + \phi g' > 0$, next period's capital. So the worker accepts μ' iff $V(\mu') > V(x')$, i.e. iff $\mu' > x_{t+1}$: the reservation match is the capital he would carry on the current job. (Under the printed law $G_x = \phi g' - \delta$ can be negative and this monotonicity argument breaks.)

First-order conditions. Let $K(x') = \int \max\{V(\mu') - V(x'), 0\} dF$ (option value of an offer). The objective is $x(1 - \phi - s) + \beta[V(x') + \pi(s)K(x')]$:

$$s : x = \beta\pi'(s)K(x'), \quad \phi : 1 = \beta g'(x\phi) V'(x') [1 - \pi(s)(1 - F(x'))].$$

Search trades current earnings for the option K ; investment pays only if the worker stays, which happens with probability $1 - \pi(s)[1 - F(x')]$. With $\pi(s) = \sqrt{s}$ the first condition gives $s = (\beta K(x')/2x)^2$ (capped by the time constraint), so only ϕ needs a grid search.

(c) Parameters (not given in the book): $A = 1$, $\alpha = .5$, $\delta = .1$, $\beta = .95$, $\mu \in \{5, 10, \dots, 45\}$ with $f_i = 1/9$, $x \in [.5, 100]$ (300 points, linear interpolation), ϕ on 400 points, $s \leq .999 - \phi$ so that $\pi(s) < 1$.

Numerical exercise — see the companion code

VFI converges in 468 iterations. V is strictly increasing. For $x \leq 12.15$ the worker spends all his time searching ($\phi = 0$, $s = .999$); search then falls fast ($s(20) = .32$) and stops once x exceeds every possible μ ($s(45) \approx 0$); investment is hump-shaped ($\phi(20) = .003$, $\phi(45) = .129$, $\phi(80) = .120$). A brute-force (ϕ, s) grid at $x = 3, 20, 60$ reproduces V within 0.2%; Monte Carlo of the policy (2 000 paths, 200 periods) gives 491.2 ± 1.2 vs. $V(5) = 489.5$ and 514.4 ± 1.1 vs. $V(30) = 513.8$.

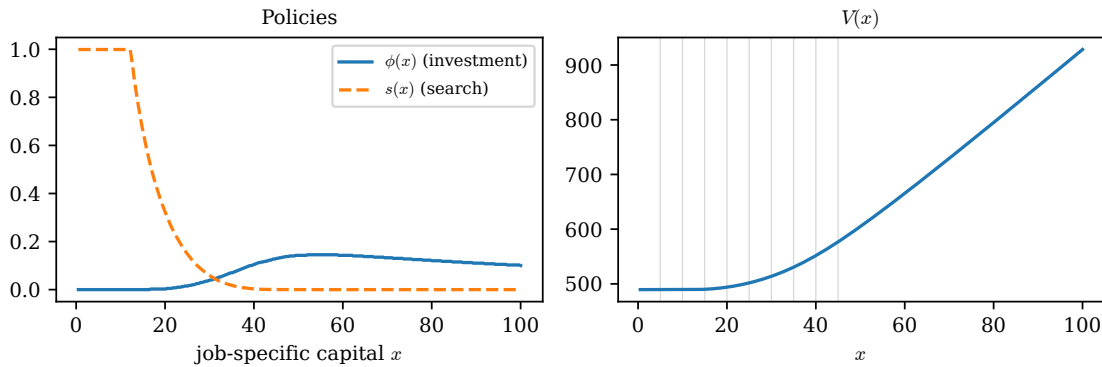


Figure 3: Exercise 6.18. Left: investment $\phi(x)$ and search $s(x)$. Right: $V(x)$; grey lines mark the possible new matches μ .

Computation: Value Iteration, Policy Improvement and Gittins Indexes

Question

(p. 217) McCall's model. (a) Bellman equation; reservation-wage policy; equation for w^* . (b) VFI: show the policy at each iteration is a reservation wage w_n ; derive a recursion; show $w_n \rightarrow w^*$ at rate β . (c) Howard's algorithm: same, and show local quadratic convergence, $w_{n+1} - w^* \cong K(w_n - w^*)^2$.

Solution:

(a) $v(w) = \max\{w/(1-\beta), c + \beta \int v dF\}$; accept iff $w \geq w^*$ with $w^* - c = \frac{\beta}{1-\beta} \int_{w^*}^B (w - w^*) dF$, equivalently $w^* = (1-\beta)c + \beta \int \max\{w, w^*\} dF$.

(b) Start from $v_0 \equiv 0$. Then $v_1(w) = \max\{w/(1-\beta), c\}$, a cutoff at $w_1 = (1-\beta)c$. If $v_n(w) = \max\{w, w_n\}/(1-\beta)$, then $v_{n+1}(w) = \max\{w/(1-\beta), c + \beta E v_n\}$ is again a cutoff rule (accept value increasing, reject value constant) with

$$w_{n+1} = T(w_n) \equiv (1-\beta)c + \beta \int \max\{w, w_n\} dF(w) = (1-\beta)c + \beta [w_n + G_F(w_n)]$$

$T'(x) = \beta[1 + G'_F(x)] = \beta F(x) \in [0, \beta]$, so by the mean value theorem $|w_{n+1} - w^*| = |T(w_n) - T(w^*)| \leq \beta |w_n - w^*|$: linear convergence at rate at most β ; asymptotically the rate is exactly $\beta F(w^*)$.

(c) *Evaluation.* The rule "accept iff $w \geq w_n$ " is worth $w/(1-\beta)$ for $w \geq w_n$ and Q_n otherwise, with $Q_n = c + \beta [F(w_n)Q_n + \frac{1}{1-\beta} \int_{w_n}^B w dF]$, so $Q_n = \frac{c + \frac{\beta}{1-\beta} \int_{w_n}^B w dF}{1 - \beta F(w_n)}$.

Improvement. Maximise $\max\{w/(1-\beta), c + \beta E v_{\text{pol}}\}$; the constant is $c + \beta E v_{\text{pol}} = Q_n$, so the new policy is again a cutoff:

$$w_{n+1} = H(w_n) \equiv \frac{(1-\beta)c + \beta \int_{w_n}^B w dF(w)}{1 - \beta F(w_n)}$$

Fixed point. $H(w^*) = w^* \iff w^*(1 - \beta F(w^*)) = (1-\beta)c + \beta \int_{w^*}^B w dF \iff w^* = (1-\beta)c + \beta \int \max\{w, w^*\} dF$, the equation of (a). *Derivative* (quotient rule, $\frac{d}{dx} \int_x^B w dF = -x f(x)$):

$$H'(x) = \frac{-\beta x f(x)[1 - \beta F(x)] + \beta f(x)[(1-\beta)c + \beta \int_x^B w dF]}{[1 - \beta F(x)]^2} = \frac{\beta f(x)[H(x) - x]}{1 - \beta F(x)},$$

so $H'(w^*) = 0$, and differentiating once more at w^* (the term multiplying $H - x$ drops out): $H''(w^*) = \frac{\beta f(w^*)[H'(w^*) - 1]}{1 - \beta F(w^*)} = -\frac{\beta f(w^*)}{1 - \beta F(w^*)}$. A second-order Taylor expansion gives

$$w_{n+1} - w^* = H(w_n) - H(w^*) \cong \frac{1}{2} H''(w^*)(w_n - w^*)^2,$$

$$K = -\frac{\beta f(w^*)}{2[1 - \beta F(w^*)]}$$

$K < 0$: after the first step Howard approaches w^* from below, doubling the number of correct digits each step. It is Newton's method on $w - T(w) = 0$: indeed $H(x) = x - \frac{x - T(x)}{1 - T'(x)}$.

Numerical exercise — see the companion code

F uniform on $[0, 1]$, $c = .3$, $\beta = .95$: $w^* = 0.7761278834$. VFI ratio $(w_{n+1} - w^*)/(w_n - w^*) \rightarrow 0.737321 = \beta F(w^*)$ (every ratio $< \beta$). Howard from $w_0 = 0$: 0.49, 0.703372, 0.768550, 0.776027, 0.776128; measured $(w_{n+1} - w^*)/(w_n - w^*)^2 = -1.760, -1.808$ vs. $K = -1.8083$. Both scalar recursions are reproduced by whole-function VFI and policy evaluation on a 4000-point grid.

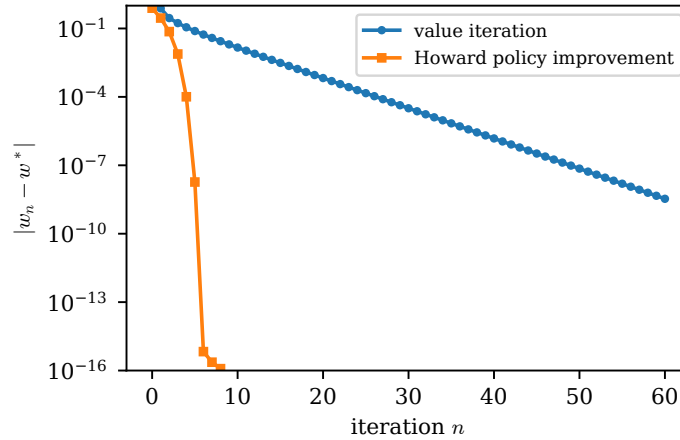


Figure 4: Exercise 6.19. Error $|w_n - w^*|$: linear for value iteration (rate $\beta F(w^*) = .737$), quadratic for Howard (machine precision after six steps).

Exercise 6.17 Gittins Indexes for Beginners

Question

(pp. 215–216) Two jobs, A and B, with wages driven by job-specific n -state Markov chains $A = [A_{ij}]$, $B = [B_{IJ}]$ and wages $w_a(i)$, $w_b(I)$. At the end of each period the worker chooses the job for next period, before seeing next period's wage; a job he returns to restarts from his last state there. No unemployment. (a) Bellman equation for a worker whose last states were i on A and I on B. (b) For a new worker whose first wage on A (B) is drawn from π_a (π_b).

Solution:

(a) Let $V(i, I)$ be the value at the end of a period with last states i and I . Working at A next period moves A's state to j with probability A_{ij} , pays $w_a(j)$, and leaves B's state frozen at I :

$$V(i, I) = \max \left\{ \sum_j A_{ij} [w_a(j) + \beta V(j, I)], \sum_J B_{IJ} [w_b(J) + \beta V(i, J)] \right\}$$

(b) Following the hint, $v_a(i)$ is the value of a worker last in state i on A who has never

worked on B (B would start from π_b); $v_b(I)$ symmetrically:

$$\begin{aligned} v_a(i) &= \max \left\{ \sum_j A_{ij}[w_a(j) + \beta v_a(j)], \sum_J \pi_{bJ}[w_b(J) + \beta V(i, J)] \right\}, \\ v_b(I) &= \max \left\{ \sum_J B_{IJ}[w_b(J) + \beta v_b(J)], \sum_i \pi_{ai}[w_a(i) + \beta V(i, I)] \right\}, \\ \text{new worker: } &\max \left\{ \sum_i \pi_{ai}[w_a(i) + \beta v_a(i)], \sum_I \pi_{bI}[w_b(I) + \beta v_b(I)] \right\}. \end{aligned}$$

Equivalently, add a "fresh" state 0 to each chain with transition row π_a (resp. π_b): then $v_a(i) = V(i, 0_b)$, $v_b(I) = V(0_a, I)$ and the new worker's value is $V(0_a, 0_b)$.

Why "Gittins". This is a two-armed bandit: the idle job's state is frozen. By the Gittins index theorem the optimal rule is to work at the job whose current state has the larger index $\nu(k) = (1 - \beta)M_k(k)$, where M_k solves the one-job "restart in k " problem $M_k(i) = \max\{r(i) + \beta \sum_j P_{ij}M_k(j), r(k) + \beta \sum_j P_{kj}M_k(j)\}$, $r(i) = \sum_j P_{ij}w(j)$. The two-dimensional Bellman equation collapses to two one-dimensional indexes.

Numerical exercise — see the companion code

$n = 4$, random (seeded) A, B , wages on $[0, 10]$, π_a, π_b ; $\beta = .9$. Indexes: A (5.858, 5.985, 6.075, 5.278), B (6.289, 5.926, 4.899, 4.595). At every (i, I) where they differ, the choice from V equals the choice of the larger index. The hint equations and the augmented-chain formulation agree; a new worker starts at B (value 55.780 vs. 55.705; fresh-state indexes 6.264 vs. 5.852).

Neal's Model of Career Choice

Exercise 6.26 Neal Model's Markov Implications

Question

(pp. 221–222) Partition the (θ, ϵ) space of Neal's model with Figure 6.5.2 into $s = 1$ (wants a new life), $s = 2$ (keeps career, wants a new job), $s = 3$ (stays put); $P_{ij} =$

$$\Pr(s_{t+1} = j \mid s_t = i), \Pr(s_0 = 1) = 1. \text{ (a) Show } P = \begin{bmatrix} 1 - P_{12} - P_{13} & P_{12} & P_{13} \\ 0 & 1 - P_{23} & P_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

and say how to compute its elements. (b) Invariant distribution(s). (c) Is the chain ergodic? (d) How could quarterly panel data on job and occupation switches be used to estimate P ?

Solution:

Facts from section 6.5. $Q = \iint [\theta' + \epsilon' + \beta v] dF dG$, $C(\theta) = \theta + \int [\epsilon' + \beta v(\theta, \epsilon')] dG$, $C(\bar{\theta}) = Q$ (6.5.4). For $\theta \geq \bar{\theta}$ the career is kept forever and the job cutoff $\bar{\epsilon}$ solves the McCall equation (6.5.5), *independent of θ* . For $\theta < \bar{\theta}$ the worker stays iff $\theta + \epsilon \geq (1 - \beta)Q$ and otherwise takes a new life. So: region 3 = $\{\theta \geq \bar{\theta}, \epsilon \geq \bar{\epsilon}\} \cup \{\theta < \bar{\theta}, \theta + \epsilon \geq (1 - \beta)Q\}$, region 2 = $\{\theta \geq \bar{\theta}, \epsilon < \bar{\epsilon}\}$, region 1 the rest.

(a) *Row 3.* A worker who stays put keeps (θ, ϵ) and stays put again: $P_{33} = 1$. *Row 2.* Here $\theta \geq \bar{\theta}$, the career is kept and only $\epsilon' \sim G$ is new; the new pair has $\theta \geq \bar{\theta}$, so it is never in region 1 ($P_{21} = 0$), and it is in region 3 iff $\epsilon' \geq \bar{\epsilon}$, whatever θ :

$$P_{23} = 1 - G(\bar{\epsilon}^-), \quad P_{22} = G(\bar{\epsilon}^-).$$

Because $\bar{\epsilon}$ does not depend on θ , row 2 is the same for every θ in region 2, which is what makes s a Markov state. *Row 1.* A fresh pair $(\theta', \epsilon') \sim F \times G$:

$$P_{12} = [1 - F(\bar{\theta}^-)] G(\bar{\epsilon}^-), \quad P_{13} = \iint \mathbf{1}\{(\theta', \epsilon') \in \text{region 3}\} dF dG,$$

$$P_{11} = 1 - P_{12} - P_{13}.$$

Computation. Solve (6.5.1) by VFI on the grid used for Figure 6.5.2; read off Q , $C(\theta)$, $\bar{\theta}$, $\bar{\epsilon}$; sum the grid probabilities of each region.

(b) $\pi'P = \pi'$: column 1 gives $\pi_1(1 - P_{12} - P_{13}) = \pi_1 \Rightarrow \pi_1 = 0$ (as $P_{12} + P_{13} > 0$); column 2 gives $\pi_1 P_{12} + \pi_2(1 - P_{23}) = \pi_2 \Rightarrow \pi_2 P_{23} = 0 \Rightarrow \pi_2 = 0$. The unique invariant distribution is $\boxed{\pi_\infty = (0, 0, 1)}$, and from $\pi_0 = (1, 0, 0)$, $\pi_t \rightarrow \pi_\infty$: everyone eventually settles in a job and career.

(c) States 1 and 2 are transient and 3 is absorbing, so the chain is not irreducible. But it has a unique stationary distribution, and every invariant function is (trivially) constant on its support $\{3\}$: in the sense of section 2.2.7 the chain *is* ergodic, and time averages of any $y(s_t)$ converge to $y(3)$ from any initial state, since absorption occurs with probability one.

(d) The state is revealed by the next move. For each worker and quarter t : a switch of job *and* occupation at $t + 1$ means $s_t = 1$; a switch of job only means $s_t = 2$; no switch means $s_t = 3$. This gives each worker's sequence $s_0 = 1, s_1, \dots$ (the last quarter is censored). The maximum-likelihood estimator of a Markov chain is the matrix of transition frequencies, with the model's zeros imposed:

$$\hat{P}_{12} = \frac{n_{12}}{n_1}, \quad \hat{P}_{13} = \frac{n_{13}}{n_1}, \quad \hat{P}_{23} = \frac{n_{23}}{n_2}, \quad \text{s.e.}(\hat{P}_{ij}) = \sqrt{\hat{P}_{ij}(1 - \hat{P}_{ij})/n_i},$$

n_{ij} the number of $i \rightarrow j$ transitions. Practical issues: a model period need not be a quarter (several switches in a quarter, or quarters without a decision); the model's prediction that a worker who has not switched never switches again (and never leaves region 2 for region 1) is testable, and violations point to learning (Jovanovic, section 6.8) or heterogeneity; with P in hand one can go further and pick F , G , β to match $\hat{P}$ (minimum distance).

Numerical exercise — see the companion code

Neal's parameters (50-point uniform grids on $[0, 5]$, $\beta = .95$): $\bar{\theta} = 3.980$, $\bar{\epsilon} = 4.184$ (as in Figure 6.5.2),

$$P = \begin{bmatrix} .7620 & .1804 & .0576 \\ 0 & .8200 & .1800 \\ 0 & 0 & 1 \end{bmatrix}.$$

Row 2 is identical for all 11 careers in region 2. A simulated panel of 5 000 workers over 40 periods, treated as in (d), gives $\hat{P}_{12} = .1813$, $\hat{P}_{13} = .0576$, $\hat{P}_{23} = .1782$, all within 4 standard errors.

Exercise 6.27 Neal Model with Unemployment**Question**

(p. 222) As in Neal's model, but a worker who wants a new job, or a new career and job, must be unemployed this period (receiving c) and draws ϵ' (or (θ', ϵ')) for next period. (a) Bellman equation for $v(\theta, \epsilon)$. (b) Characterise the optimal policy.

Solution:

(a)

$$v(\theta, \epsilon) = \max \left\{ \theta + \epsilon + \beta v(\theta, \epsilon), \quad c + \beta \int v(\theta, \epsilon') dG(\epsilon'), \right. \\ \left. c + \beta \iint v(\theta', \epsilon') dF(\theta') dG(\epsilon') \right\}$$

(b) Staying put, once optimal, is optimal forever and is worth $(\theta + \epsilon)/(1 - \beta)$. Let $C(\theta) = c + \beta \int v(\theta, \epsilon') dG$ and $Q = c + \beta \iint v dF dG$. v is nondecreasing in θ (by induction on VFI), so C is nondecreasing and there is a cutoff $\bar{\theta}$ with $C(\theta) \geq Q$ iff $\theta \geq \bar{\theta}$.

- $\theta < \bar{\theta}$: a new job is dominated by a new life. Stay iff $\theta + \epsilon \geq (1 - \beta)Q$ (a diagonal boundary), otherwise new life.
- $\theta \geq \bar{\theta}$: the career is never abandoned (a new job leaves θ unchanged). The problem is McCall with offers $\theta + \epsilon'$ and compensation c : $v(\theta, \epsilon) = \max\{\frac{\theta + \epsilon}{1 - \beta}, c + \beta \int v(\theta, \epsilon') dG\}$. Stay iff $\epsilon \geq \bar{\epsilon}(\theta)$ with

$$\theta + \bar{\epsilon}(\theta) - c = \frac{\beta}{1 - \beta} \int_{\bar{\epsilon}(\theta)} (\epsilon' - \bar{\epsilon}(\theta)) dG(\epsilon'), \quad \frac{d\bar{\epsilon}}{d\theta} = -\frac{1}{1 + \frac{\beta}{1 - \beta}[1 - G(\bar{\epsilon})]} \in (-1, 0).$$

Contrast with Neal. There $\bar{\epsilon}$ is independent of θ because a new job is drawn at once and the career income θ is earned regardless. Here looking for a new job costs a period of career income (θ replaced by c), so workers in better careers are less choosy about jobs: $\bar{\epsilon}(\theta)$ slopes down, while $\theta + \bar{\epsilon}(\theta)$ still rises. The three regions keep the shape of Figure 6.5.2, but the new-job/stay boundary now slopes down.

Numerical exercise — see the companion code

Same grids, $\beta = .95$, $c = 1$: $\bar{\theta} = 3.673$ (vs. 3.980 without unemployment); $\bar{\epsilon}(\theta)$ falls from 3.265 at $\theta = 3.673$ to 3.098 at $\theta = 5$. For every $\theta \geq \bar{\theta}$ the VFI stay region coincides with $\epsilon \geq \bar{\epsilon}(\theta)$ from the McCall equation, and for $\theta < \bar{\theta}$ with $\theta + \epsilon \geq (1 - \beta)Q$.

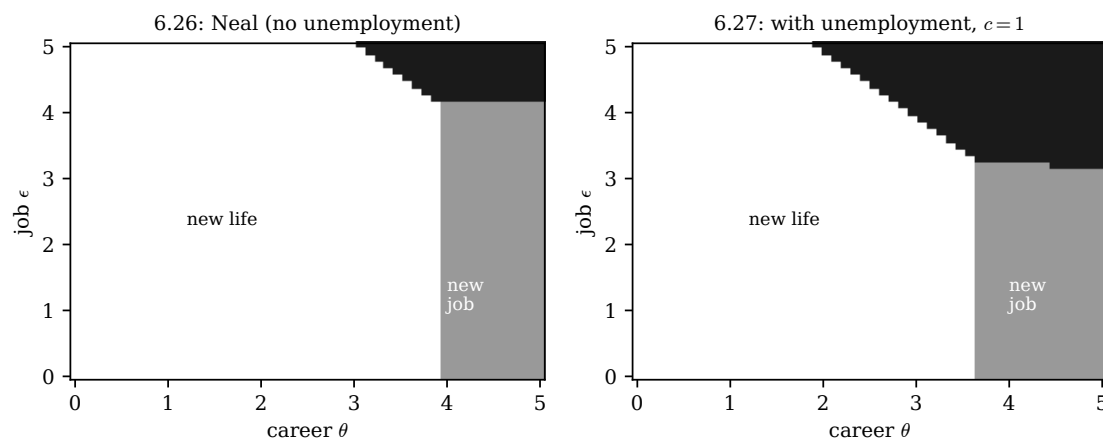


Figure 5: Exercises 6.26–6.27. Decision regions: white = new life, grey = new job, black = stay put. Left: Neal’s model (reproduces Figure 6.5.2). Right: with unemployment while searching, $c = 1$.

Book misprints and caveats

- **Exercise 6.18, law of motion.** Printed $x_{t+1} = g(x_t \phi_t) - \delta x_t$ gives negative capital for small ϕ_t ; read as $x_{t+1} = (1 - \delta)x_t + g(x_t \phi_t)$.
- **Exercise 6.8.** "The economy with the higher growth rate has the smaller reservation wage" holds strictly only for $c > 0$; with $c = 0$ the reservation wage does not depend on ϕ .
- **Exercise 6.19(b).** "Converges at rate β " is a bound; the exact asymptotic rate is $\beta F(w^*) < \beta$.
- **Exercise 6.4.** π_{mn} is indexed (next, current), so $[\pi_{mn}]$ is column-stochastic.
- **Exercise 6.27(a).** "about to decide whether to decide whether" (duplicated phrase).

Companion code

Notebook: `Resolucao/ljungqvist_sargent/ls_ch06.ipynb`.

It reproduces every numerical result above (Python: `numpy`, `scipy`, `matplotlib`), executed end to end with outputs saved. Each section checks its numbers by an independent route with `assert`.

Item	What it does
helpers	<code>resw</code> (reservation wage from (6.3.3) by bracketing), <code>mccall_vfi</code> (whole-function VFI), <code>cutoff_value</code> (exact value of a cutoff rule), <code>max_dist</code> (best of n draws).
§ 6.1–6.5, 6.7, 6.8, 6.12, 6.13	Reservation equation vs. VFI (on pairs of offers, on (w, n) , on (w, ϵ)), exhaustive search over (number of offers, cutoff), simulated durations; asset model by VFI plus exact policy evaluation.
§ 6.9, 6.10, 6.20, 6.21	Brute-force cutoffs, backward induction on functions, MPS checks, atoms, hazards; reservation price by Monte Carlo.
§ 6.14–6.16, 6.22–6.25	Multi-function VFI vs. closed forms, simulated hazards and quit rates, Howard with linear policy evaluation, enumeration of all 2^5 policies.
§ 6.6, 6.11, 6.18	Nash vs. planner search, literal VFI for Pissarides, Jovanovic by VFI with brute-force and Monte Carlo checks.
§ 6.17, 6.19	Gittins indexes (restart-in-state) vs. the two-job Bellman equation; VFI vs. Howard rates.
§ 6.26, 6.27	Neal's model and its unemployment variant; P vs. a simulated panel.
<code>fig/ls06_*.pdf</code>	The five figures used in this file.